MIJ TECHNOLOGIES • PYTHON PROGRAMMING SERIES

ASSIGNMENT 5

Hospital Patient Billing System

TOTAL MARKS	DIFFICULTY	ESTIMATED TIME	TOPICS COVERED
30 Marks	Very Hard	6 - 8 hours	Functions, Loops & Conditionals

DESIGN REQUIREMENTS: *Your solution must demonstrate structured programming with proper use of functions, loops, and conditionals. Clearly separate responsibilities into different functions — avoid writing everything in one long block of code. Validate all input (insurance status vs. provider) and use meaningful variable names. Your program must run without crashing.*

OVERVIEW

Write a Command Line Program that simulates a Hospital Patient Billing System. Your program should allow a receptionist to manage multiple patient visits, calculate medical bills based on service type and insurance coverage, and generate daily hospital revenue reports.

PART A — PATIENT INFORMATION INPUT — 6 MARKS

The program should repeatedly prompt for:

- Patient name
- Service type (Consultation, Emergency, Surgery, Admission)
- Number of days (for Admission) or hours (for other services)
- Insurance status (Yes or No)
- Insurance provider (if applicable: NHIF, AAR, Jubilee, None)

Validation: if insurance status is “No”, provider must be “None”.

PART B — SERVICE PRICING — 6 MARKS

Service	Pricing Structure
Consultation (per hour)	First hour: 50,000 UGX. Each additional hour: 30,000 UGX
Emergency (per hour)	First 2 hours: 100,000 UGX/hr. Additional hours: 80,000 UGX/hr
Surgery (flat rate)	Minor (< 3 hrs): 500,000 UGX. Major (≥ 3 hrs): 1,200,000 UGX
Admission (per day)	Days 1-3: 150,000/day. Days 4-7: 120,000/day. Beyond 7: 100,000/day

PART C — INSURANCE COVERAGE CALCULATION — 6 MARKS

Provider	Coverage	Maximum Cap
NHIF	60% of total bill	300,000 UGX

Provider	Coverage	Maximum Cap
AAR	80% of total bill	No limit
Jubilee	70% of total bill	500,000 UGX
None	0% (patient pays 100%)	N/A

```
Insurance Coverage = min(Total Bill × Coverage %, Maximum Cap)
Patient Pays = Total Bill - Insurance Coverage
```

PART D — PATIENT BILL DISPLAY — 6 MARKS

Display for each patient:

- Patient name, service type, duration
- Base service charge
- Insurance provider, coverage amount
- Patient amount (what the patient pays)

PART E — DAILY HOSPITAL REPORT — 6 MARKS

Display:

- Total patients served
- Total revenue from patients
- Average bill per patient
- Service breakdown (4 types)
- Most common service
- Insurance statistics

TEST CASES FOR VERIFICATION

Consultation, 1 hour, No Insurance

```
Service Charge : 50,000 UGX
Insurance : 0 UGX
Patient Pays : 50,000 UGX
```

Admission, 10 days, NHIF

```
Days 1-3 : 3 x 150,000 = 450,000
Days 4-7 : 4 x 120,000 = 480,000
Days 8-10: 3 x 100,000 = 300,000
TOTAL BILL = 1,230,000 UGX

NHIF (60%): 1,230,000 x 0.60 = 738,000 -> CAPPED at 300,000
Patient Pays: 1,230,000 - 300,000 = 930,000 UGX
```

Surgery, 4 hours, AAR

```
Duration >= 3 hrs -> Major Surgery = 1,200,000 UGX
AAR (80%): 1,200,000 x 0.80 = 960,000 UGX (no cap)
Patient Pays: 1,200,000 - 960,000 = 240,000 UGX
```

SAMPLE TERMINAL OUTPUT

Present your program's output using clean, full-width section separators with labelled headers, exactly as shown below.

● **TERMINAL — PART 1 OF 2**

```
================================================================================
   HOSPITAL PATIENT BILLING SYSTEM v1.0
================================================================================
Enter Patient Name: Alice Nambi
Enter Service Type (Consultation/Emergency/Surgery/Admission): Consultation
Enter Hours: 3
Insurance Status (Yes/No): Yes
Insurance Provider (NHIF/AAR/Jubilee/None): AAR

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  First hour            :              50,000 UGX
  Additional 2 hours     : 2 x 30,000 =  60,000 UGX
  Total Service Charge     :            110,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider   : AAR (80% coverage, no cap)
  Coverage   : 110,000 x 0.80 = 88,000 UGX
  Patient Pays: 110,000 - 88,000 = 22,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient  : Alice Nambi
  Service  : Consultation
  Duration : 3 hours
  Total Service Charge: 110,000 UGX
  Insurance Coverage  :  88,000 UGX
  Patient Pays        :  22,000 UGX
================================================================================
Add another patient? (yes/no): yes
```

● TERMINAL — PART 2 OF 2

```
================================================================================
Enter Patient Name: John Okello
Enter Service Type (Consultation/Emergency/Surgery/Admission): Surgery
Enter Hours: 2
Insurance Status (Yes/No): No
Insurance Provider (NHIF/AAR/Jubilee/None): None

================================================================================
   SERVICE CHARGE BREAKDOWN
================================================================================
  Duration < 3 hrs -> Minor Surgery (flat rate)
  Service Charge: 500,000 UGX

================================================================================
   INSURANCE COVERAGE
================================================================================
  Provider   : None (0% coverage)
  Coverage   : 0 UGX
  Patient Pays: 500,000 UGX

================================================================================
   PATIENT BILL
================================================================================
  Patient  : John Okello
  Service  : Surgery (Minor)
  Duration : 2 hours
  Total Service Charge: 500,000 UGX
  Insurance Coverage  :       0 UGX
  Patient Pays        : 500,000 UGX
================================================================================
Add another patient? (yes/no): no

================================================================================
   DAILY HOSPITAL REPORT
================================================================================
  Total Patients Served     : 2
  Total Revenue Collected     : 522,000 UGX
  Average Bill Per Patient      : 261,000 UGX
  Service Breakdown               : Consultation: 1 | Emergency: 0 |
                                    Surgery: 1 | Admission: 0
  Most Common Service             : Consultation / Surgery (Tie)
  Insurance Statistics              : With Insurance: 1 | Without: 1
  Most Common Provider               : AAR
================================================================================
```

MARKING SCHEME

Part	Marks	Description
A	6	Input collection with insurance validation
B	6	All service pricing tiers implemented
C	6	Insurance coverage with caps (NHIF/Jubilee)
D	6	Complete bill with coverage breakdown
E	6	Summary with service + insurance statistics
TOTAL	**30**	**100% requires every part working end-to-end with correct calculations**

100% is only awarded if all parts are completed without errors — master complex tiered pricing and insurance cap calculations!